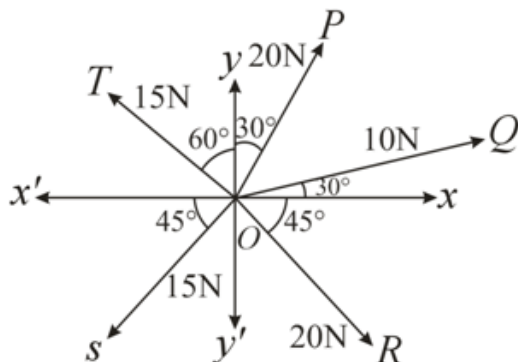


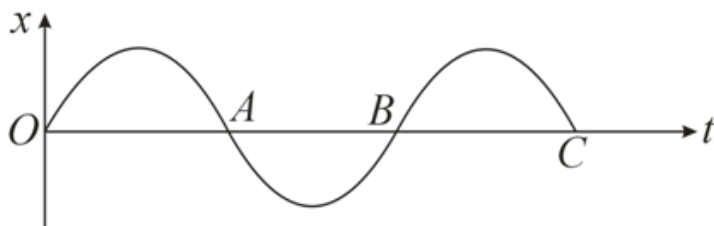
Question Paper

- Q1. The resultant of these forces $\vec{OP}$, $\vec{OQ}$, $\vec{OR}$, $\vec{OS}$ and $\vec{OT}$ is approximately _____ N.
[Take $\sqrt{3} = 1.7$, $\sqrt{2} = 1.4$ Given $\hat{i}$ and $\hat{j}$ unit vectors along x , y axis]

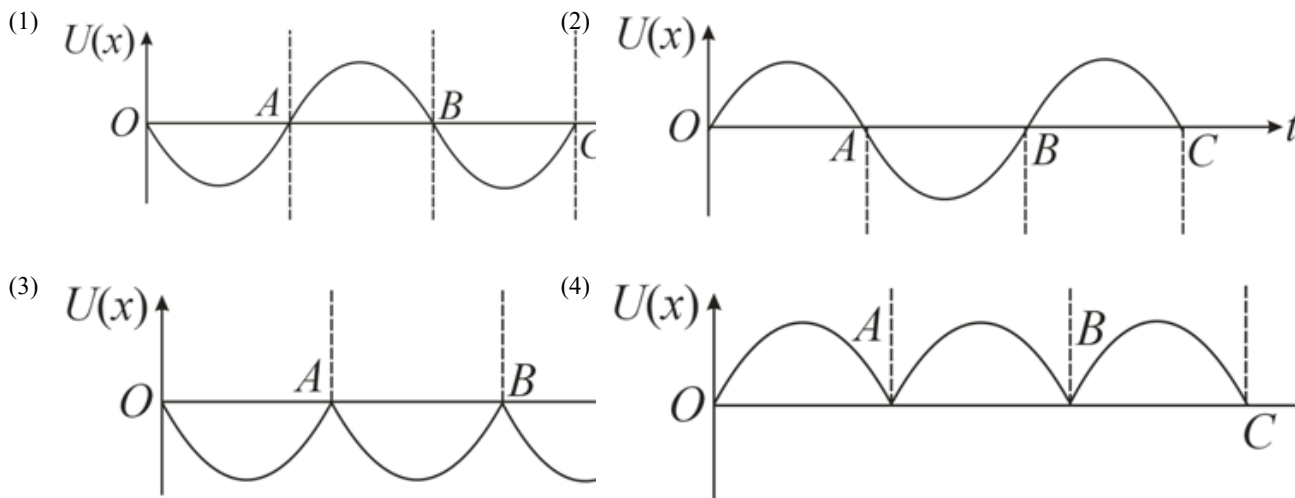


- (1) $-1.5\hat{i} - 15.5\hat{j}$ (2) $9.25\hat{i} + 5\hat{j}$
(3) $3\hat{i} + 15\hat{j}$ (4) $2.5\hat{i} - 14.5\hat{j}$
- Q2. If E and H represents the intensity of electric field and magnetizing field respectively, then the unit of $\frac{E}{H}$ will be:
(1) joule (2) ohm
(3) newton (4) mho
- Q3. Which of the following is not a dimensionless quantity?
(1) Power factor (2) Quality factor
(3) Permeability of free space (μ_0) (4) Relative magnetic permeability (μ_r)
- Q4. A huge circular arc of length 4.4 ly subtends an angle $4s$ at the centre of the circle. How long it would take for a body to complete 4 revolution if its speed is 8 AU per second?
Given : 1 ly = 9.46×10^{15} m
1 AU = 1.5×10^{11} m
(1) 3.5×10^6 s (2) 4.5×10^{10} s
(3) 4.1×10^8 s (4) 7.2×10^8 s
- Q5. Moment of inertia of a square plate of side l about the axis passing through one of the corner and perpendicular to the plane of square plate is given by:
(1) $\frac{Ml^2}{6}$ (2) $\frac{2}{3}Ml^2$
(3) Ml^2 (4) $\frac{Ml^2}{12}$
- Q6. In Millikan's oil drop experiment, what is viscous force acting on an uncharged drop of radius 2.0×10^{-5} m and density 1.2×10^3 kg m $^{-3}$? Take viscosity of liquid = 1.8×10^{-5} N s m $^{-2}$. (Neglect buoyancy due to air).
(1) 5.8×10^{-10} N (2) 3.9×10^{-10} N
(3) 1.8×10^{-10} N (4) 3.8×10^{-11} N
- Q7. An ideal gas is expanding such that $PT^3 = \text{constant}$. The coefficient of volume expansion of the gas is:
(1) $\frac{2}{T}$ (2) $\frac{3}{T}$
(3) $\frac{1}{T}$ (4) $\frac{4}{T}$
- Q8. A balloon carries a total load of 185 kg at normal pressure and temperature of 27°C . What load will the balloon carry on rising to a height at which the barometric pressure is 45 cm of Hg and the temperature is -7°C . Assuming the volume constant?
(1) 214.15 kg (2) 123.54 kg
(3) 219.07 kg (4) 181.46 kg

Q9. The variation of displacement with time of a particle executing free simple harmonic motion is shown in the figure.



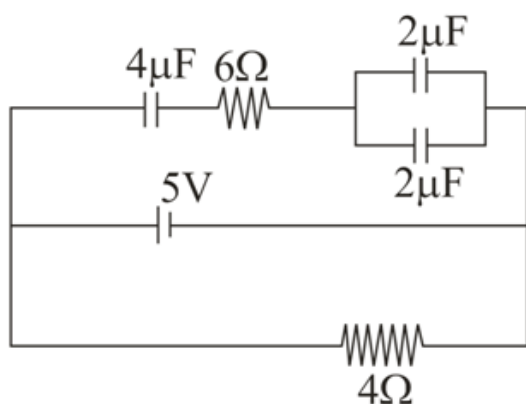
The potential energy $U(x)$ versus time (t) plot of the particle is correctly shown in figure:



Q10. A uniformly charged disc of radius R having surface charge density σ is placed in the xy plane with its center at the origin. Find the electric field intensity along the z -axis at a distance Z from origin:

- (1) $E = \frac{2\epsilon_0}{\sigma} \left(\frac{1}{(Z^2 + R^2)^{1/2}} + Z \right)$ (2) $E = \frac{\sigma}{2\epsilon_0} \left(1 + \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$
 (3) $E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{Z}{(Z^2 + R^2)^{1/2}} \right)$ (4) $E = \frac{\sigma}{2\epsilon_0} \left(\frac{1}{(Z^2 + R^2)} + \frac{1}{Z^2} \right)$

Q11. Calculate the amount of charge on capacitor of $4 \mu\text{F}$. The internal resistance of battery is 1Ω :



- (1) $4 \mu\text{C}$ (2) $8 \mu\text{C}$
 (3) $16 \mu\text{C}$ (4) zero

Q12. Five identical cells each of internal resistance 1Ω and emf 5 V are connected in series and in parallel with an external resistance R . For what value of R , current in series and parallel combination will remain the same?

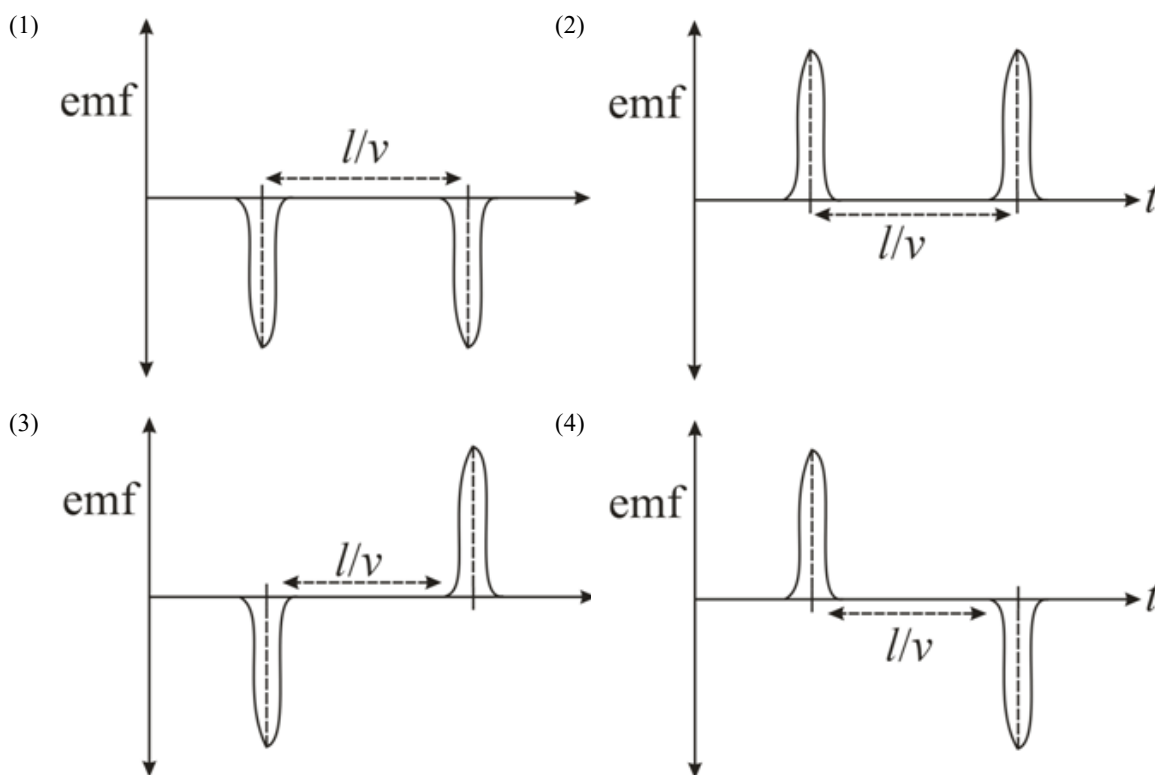
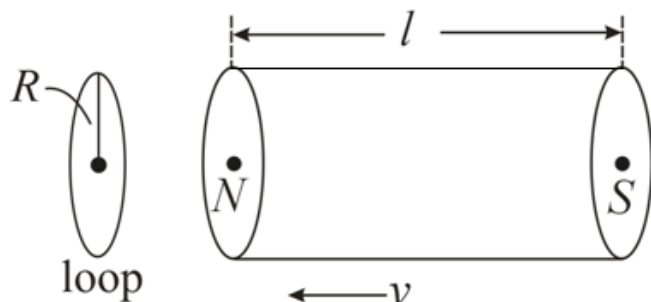
- (1) 1Ω (2) 5Ω
 (3) 25Ω (4) 10Ω

Question Paper

Q13. Two ions of masses 4 amu and 16 amu have charges $+2e$ and $+3e$ respectively. These ions pass through the region of the constant perpendicular magnetic field. The kinetic energy of both ions is the same. Then:

- (1) lighter ion will be deflected less than heavier ion (2) lighter ion will be deflected more than heavier ion
(3) both ions will be deflected equally (4) no ion will be deflected

Q14. A bar magnet is passing through a conducting loop of radius R with velocity v . The radius of the bar magnet is such that it just passes through the loop. The induced e.m.f. in the loop can be represented by the approximate curve:



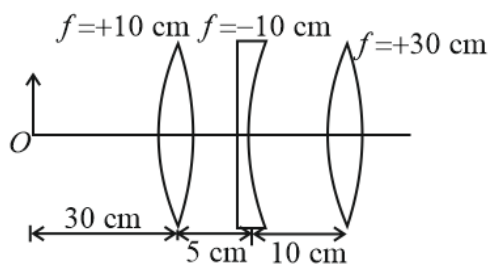
Q15. The electric field in a plane electromagnetic wave is given by, $E = 50 \sin(500x - 10 \times 10^{10}t)$ V m $^{-1}$. The velocity of an electromagnetic wave in this medium is: (Given c = the speed of light in vacuum).

- (1) $\frac{3}{2}c$ (2) c
(3) $\frac{c}{2}$ (4) $\frac{2}{3}c$

Q16. An object is placed beyond the centre of curvature C of the given concave mirror. If the distance of the object is d_1 from C and the distance of the image formed is d_2 from C , the radius of curvature of this mirror is:

- (1) $\frac{2d_1d_2}{d_1+d_2}$ (2) $\frac{2d_1d_2}{d_1-d_2}$
(3) $\frac{d_1d_2}{d_1+d_2}$ (4) $\frac{d_1d_2}{d_1-d_2}$

Q17. Find the distance of the image from object O , formed by the combination of lenses in the figure:



- (1) 75 cm (2) 10 cm
(3) infinity (4) 20 cm

Q18. In a photoelectric experiment, increasing the intensity of incident light:

- (1) increases the number of photons incident and also increases the K.E. of the ejected electrons.
(2) increases the frequency of photons incident and increases the K.E. of the ejected electrons.
(3) increases the number of photons incident and the K.E. of the ejected electrons remains unchanged.
(4) increases the frequency of photons incident and the K.E. of the ejected electrons remains unchanged.

Q19. There are 10^{10} radioactive nuclei in a given radioactive element. Its half-life time is 1 min. How many nuclei will remain after 30 s? ($\sqrt{2} = 1.414$)

- (1) 7×10^9 (2) 10^5
(3) 2×10^{10} (4) 4×10^{10}

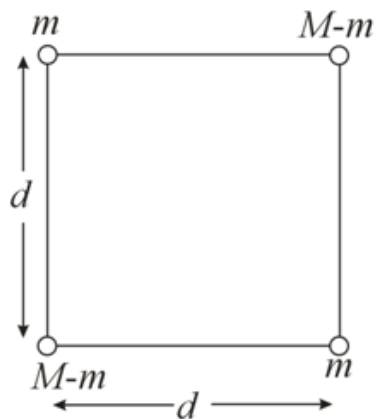
Q20. For a transistor in CE mode to be used as an amplifier, it must be operated in:

- (1) Cut-off region only (2) Saturation region only
(3) The active region only (4) Both cut-off and Saturation

Q21. If the velocity of a body related to displacement x is given by $v = \sqrt{5000 + 24x} \text{ m s}^{-1}$, then the acceleration of the body is _____ m s^{-2} .

Q22. Two persons A and B perform same amount of work in moving a body through a certain distance d with application of forces acting at angles 45° and 60° with the direction of displacement respectively. The ratio of force applied by person A to the force applied by person B is $\frac{1}{\sqrt{x}}$. The value of x is _____.

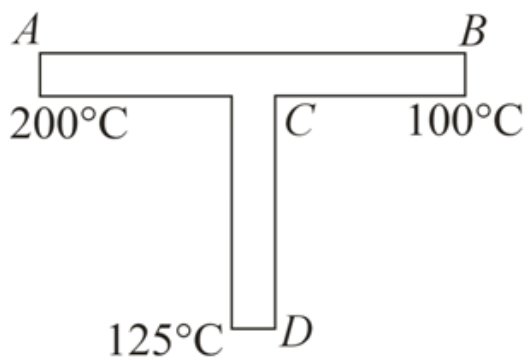
Q23. A body of mass $(2M)$ splits into four masses $\{m, M - m, m, M - m\}$, which are rearranged to form a square as shown in the figure. The ratio of $\frac{M}{m}$ for which, the gravitational potential energy of the system becomes maximum is $x : 1$. The value of x is _____.



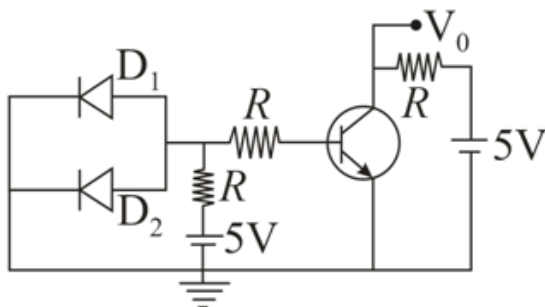
Q24. A rod CD of thermal resistance 10.0 KW^{-1} is joined at the middle of an identical rod AB as shown in figure. The ends A , B and D are maintained at 200°C , 100°C and 125°C respectively. The heat current in CD is

Question Paper

P W. The value of P is



- Q25.** Two cars X and Y are approaching each other with velocities 36 km h^{-1} and 72 km h^{-1} respectively. The frequency of a whistle sound as emitted by a passenger in car X , heard by the passenger in car Y is 1320 Hz . If the velocity of sound in air is 340 ms^{-1} , the actual frequency of the whistle sound produced is _____ Hz .
- Q26.** First, a set of n equal resistors of 10Ω each are connected in series to a battery of E.M.F. 20 V and internal resistance 10Ω . A current I is observed to flow. Then, the n resistors are connected in parallel to the same battery. It is observed that the current is increased 20 times, then the value of n is _____.
- Q27.** A uniform conducting wire of length is $24a$, and resistance R is wound up as a current carrying coil in the shape of an equilateral triangle of side a and then in the form of a square of side a . The coil is connected to a voltage source V_0 . The ratio of magnetic moment of the coils in case of equilateral triangle to that for square is $1 : \sqrt{y}$ where y is _____.
- Q28.** The alternating current is given by, $i = \left\{ \sqrt{42} \sin\left(\frac{2\pi}{T}t\right) + 10 \right\} \text{ A}$. The R.M.S. value of this current is _____ A .
- Q29.** A circuit is arranged as shown in figure. The output voltage V_o is equal to _____ V .



- Q30.** A transmitting antenna has a height of 320 m and that of receiving antenna is 2000 m . The maximum distance between them for satisfactory communication in line of sight mode is d . The value of d is _____ km .
- Q31.** The unit of the van der Waals gas equation parameter a in $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ is:
- | | |
|--|-------------------------|
| (1) $\text{dm}^3 \text{ mol}^{-1}$ | (2) kg ms^{-1} |
| (3) $\text{atm dm}^6 \text{ mol}^{-2}$ | (4) kg ms^{-2} |
- Q32.** In polythionic acid, $\text{H}_2\text{S}_x\text{O}_6$ ($x = 3$ to 5) the oxidation state(s) of sulphur is/are:
- | | |
|-----------------------|------------------------|
| (1) $+6$ only | (2) $+5$ only |
| (3) 0 and $+5$ only | (4) $+3$ and $+5$ only |

Q33. Deuterium resembles hydrogen in properties but:

Question Paper

- (1) reacts vigorously than hydrogen
(3) emits β^+ particles

- (2) reacts just as hydrogen
(4) reacts slower than hydrogen

Q34. The number of water molecules in gypsum, dead burnt plaster and plaster of Paris, respectively are:

- (1) 2, 0 and 0.5
(3) 2, 0 and 1

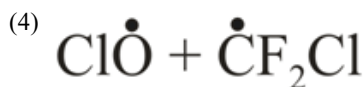
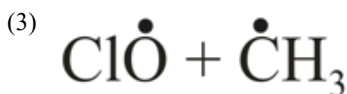
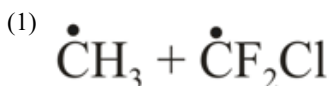
- (2) 0.5, 0 and 2
(4) 5, 0 and 0.5

Q35. In which one of the following molecules strongest back donation of an electron pair from halide to boron is expected?

- (1) BI_3
(3) BCl_3

- (2) BBr_3
(4) BF_3

Q36. The gas /A/ is having very low reactivity reaches to stratosphere. It is non-toxic and non-flammable but dissociated by UV-radiations in stratosphere. The intermediates formed initially from the gas /A/ are:



Q37. Match List - I with List - II:

List-I (Property)

List-II (Example)

- (a) Diamagnetism
(b) Ferrimagnetism
(c) Paramagnetism
(d) Antiferromagnetism

- (i) MnO
(ii) O_2
(iii) NaCl
(iv) Fe_3O_4

Choose the most appropriate answer from the options given below:

- (1) (a) – (iv), (b) – (ii), (c) – (i), (d) – (iii) (2) (a) – (ii), (b) – (i), (c) – (iii), (d) – (iv)
(3) (a) – (i), (b) – (iii), (c) – (iv), (d) – (ii) (4) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

Q38. Tyndall effect is more effectively shown by:

- (1) lyophilic colloid
(3) lyophobic colloid

- (2) suspension
(4) true solution

Q39. Which refining process is generally used in the purification of low melting metals?

- (1) Chromatographic method
(3) Zone refining

- (2) Electrolysis
(4) Liquation

Q40. Match List - I with List - II:

List-I (Species)

List-II (Number of lone pairs of electrons on the central atom)

- (a) XeF_2 (i)
(b) XeO_2F_2 (ii)
(c) XeO_3F_2 (iii)
(d) XeF_4 (iv)

- 0
1
2
3

Choose the most appropriate answer from the options given below:

- (1) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i) (2) (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii)
(3) (a) – (iii), (b) – (ii), (c) – (iv), (d) – (i) (4) (a) – (iv), (b) – (ii), (c) – (i), (d) – (iii)

Q41. The nature of oxides V_2O_3 and CrO is indexed as /X/ and /Y/ type respectively. The correct set of X and Y is:

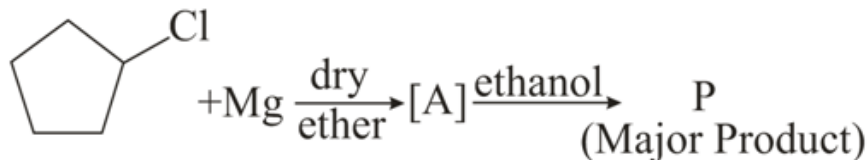
- (1) X =acidic Y =acidic
(3) X =basic Y =basic

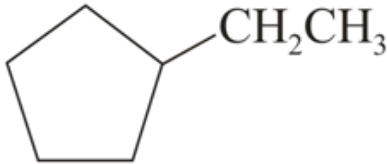
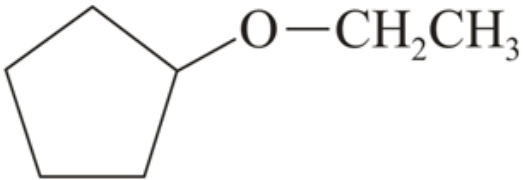
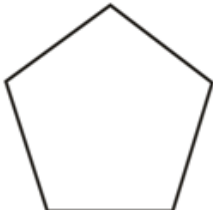
- (2) X =basic Y =amphoteric
(4) X =amphoteric Y =basic

Q42. Acidic ferric chloride solution on treatment with excess of potassium ferrocyanide gives a Prussian blue coloured colloidal species. It is:

- (1) $\text{HFe}[\text{Fe}(\text{CN})_6]$ (2) $\text{K}_5\text{Fe}[\text{Fe}(\text{CN})_6]_2$
(3) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (4) $\text{KFe}[\text{Fe}(\text{CN})_6]$

Q43. In the following sequence of reactions the P is:



- (1)  (2) 
(3)  (4) 

Q44. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

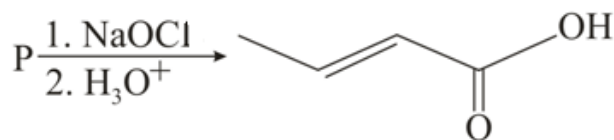
Assertion (A) : Synthesis of ethyl phenyl ether may be achieved by Williamson synthesis.

Reason (R): Reaction of bromobenzene with sodium ethoxide yields ethyl phenyl ether.

In the light of the above statements, choose the most appropriate answer from the options given below.

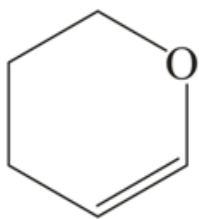
- (1) Both (A) and (R) are correct but (R) is NOT the correct explanation of (A) (2) (A) is correct but (R) is not correct
(3) (A) is not correct but (R) is correct (4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

Q45. The structure of the starting compound P used in the reaction given below is:

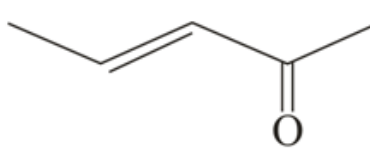


Question Paper

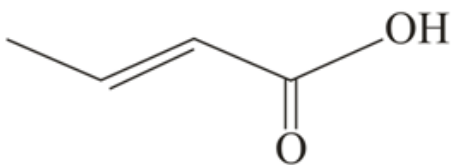
(1)



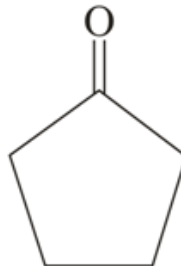
(2)



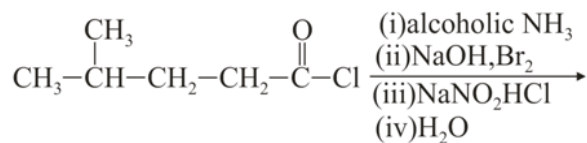
(3)



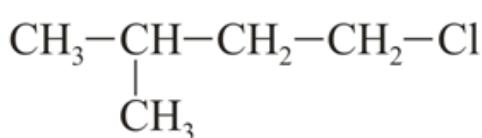
(4)



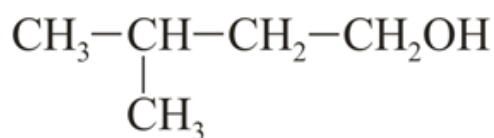
Q46. The major product of the following reaction is:



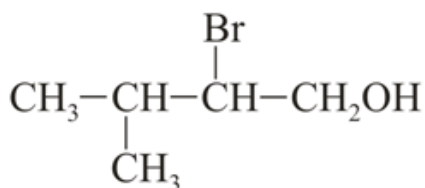
(1)



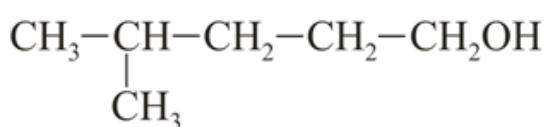
(2)



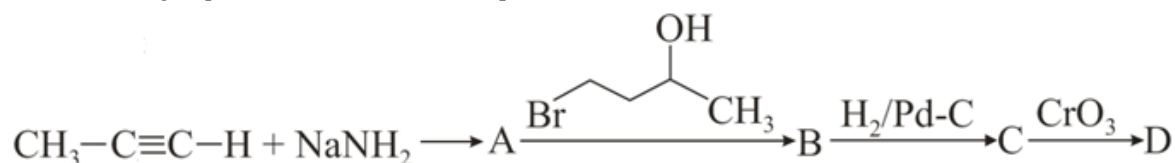
(3)



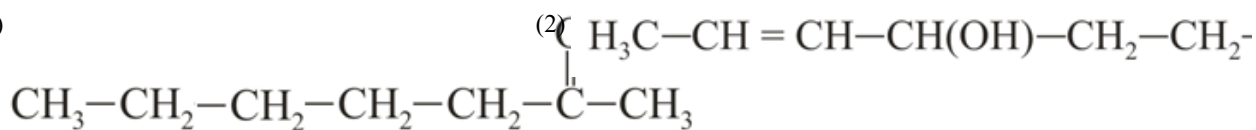
(4)



Q47. In the following sequence of reactions, the final product D is:



(1)



(2)



(3)



(4)

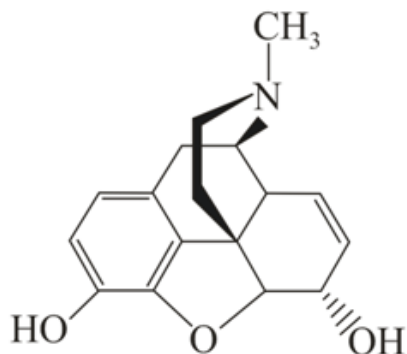


Q48. Which of the following is not a correct statement for primary aliphatic amines?

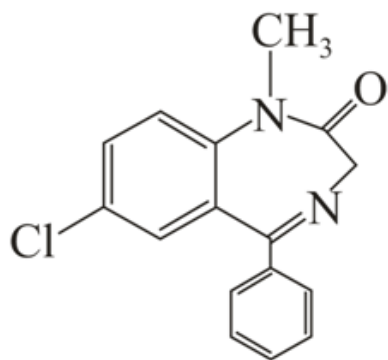
Question Paper

- (1) Primary amines can be prepared by the Gabriel phthalimide synthesis. (2) Primary amines are less basic than the secondary amines.
- (3) The intermolecular association in primary amines is less than the intermolecular association in secondary amines. (4) Primary amines on treating with nitrous acid solution form corresponding alcohols except methyl amine.

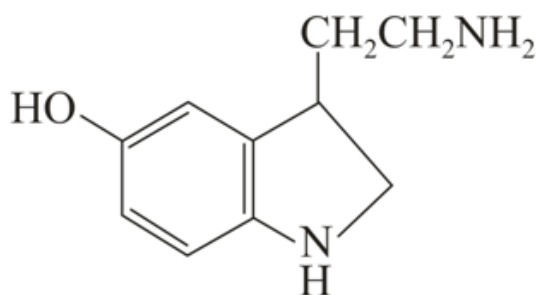
Q49.



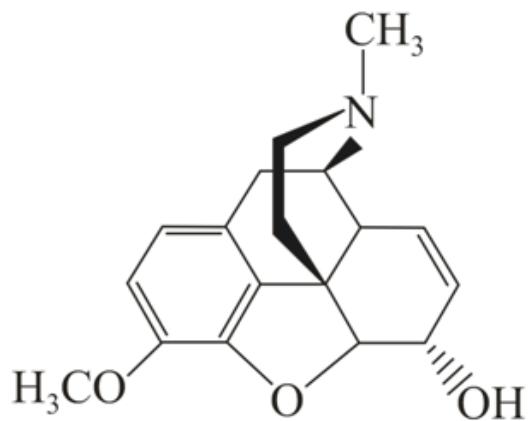
A



B



C



D

The correct statement about (A), (B), (C) and (D) is:

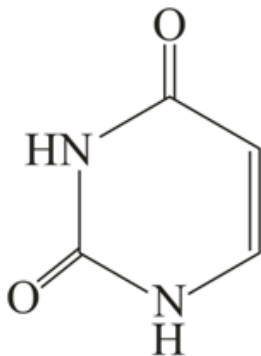
Question Paper

- (1) (B) and (C) are tranquillizers
 (3) (A) and (D) are tranquillizers

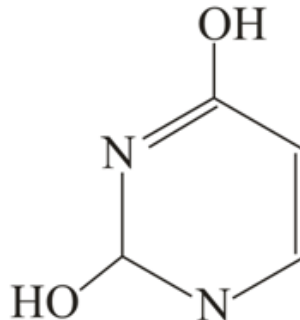
- (2) (B), (C) and (D) are tranquillizers
 (4) (A), (B) and (C) are narcotic analgesics

Q50. Out of following isomeric forms of uracil, which one is present in RNA?

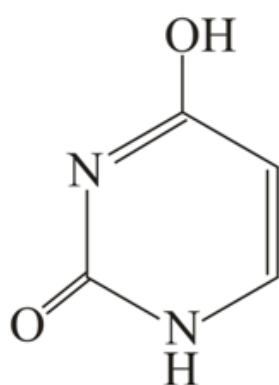
(1)



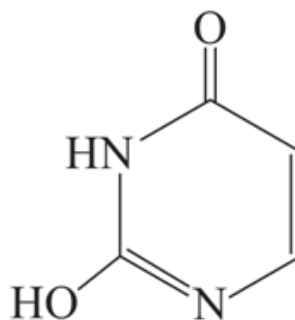
(2)



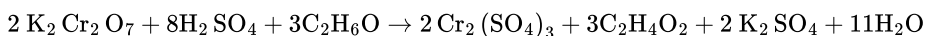
(3)



(4)



Q51. The reaction that occurs in a breath analyser, a device used to determine the alcohol level in a person's blood stream is



If the rate of appearance of $\text{Cr}_2(\text{SO}_4)_3$ is $2.67 \text{ mol min}^{-1}$ at a particular time, the rate of disappearance of $\text{C}_2\text{H}_6\text{O}$ at the same time is mol min^{-1} . (Nearest integer)

Q52. 1 kg of 0.75 molal aqueous solution of sucrose can be cooled up to -4°C before freezing. The amount of ice (in g) that will be separated out is _____. (Nearest integer) [Given : $K_f(\text{H}_2\text{O}) = 1.86 \text{ K kg mol}^{-1}$]

Q53. The number of f electrons in the ground state electronic configuration of $\text{Np}(Z = 93)$ is _____. (Integer answer)

Q54. The number of moles of NH_3 , that must be added to 2 L of 0.80 M AgNO_3 in order to reduce the concentration of Ag^+ ions to $5.0 \times 10^{-8} \text{ M}$ ($K_{\text{formation}}$ for $[\text{Ag}(\text{NH}_3)_2]^+ = 1.0 \times 10^8$) is _____. (Nearest integer) [Assume no volume change on adding NH_3]

Q55. When 10 mL of an aqueous solution of KMnO_4 was titrated in acidic medium, equal volume of 0.1 M of an aqueous solution of ferrous sulphate was required for complete discharge of colour. The strength of KMnO_4 in grams per litre is $\times 10^{-2}$. (Nearest integer) [Atomic mass of $\text{K} = 39$, $\text{Mn} = 55$, $\text{O} = 16$]

Q56. The number of moles of CuO , that will be utilized in Dumas method for estimating nitrogen in a sample of 57.5 g of N, N-dimethylaminopentane is $\times 10^{-2}$. (Nearest integer)

Q57. 200 mL of 0.2 M HCl is mixed with 300 mL of 0.1 M NaOH . The molar heat of neutralization of this reaction is -57.1 kJ . The increase in temperature in $^\circ\text{C}$ of the system on mixing is $x \times 10^{-2}$. The value of x is (Nearest integer)

[Given: Specific heat of water = $4.18 \text{ J g}^{-1} \text{ K}^{-1}$]

Question Paper

Density of water = 1.00 g cm^{-3}

(Assume no volume change on mixing)

Q58. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is equal to $\frac{h^2}{x m a_0^2}$. The value of

$10x$ is (a_0 is radius of Bohr's orbit)

(Nearest integer)

[Given: $\pi = 3.14$]

Q59. In Carius method for estimation of halogens, 0.2 g of an organic compound gave 0.188 g of AgBr. The percentage of bromine in the compound is _____. (Nearest integer)

[Atomic mass: Ag = 108, Br = 80]

Q60. 1 mol of an octahedral metal complex with formula $\text{MCl}_3 \cdot 2\text{L}$ on reaction with excess of AgNO_3 gives 1 mol of AgCl. The denticity of Ligand L is _____. (Integer answer)

Q61. If $x^2 + 9y^2 - 4x + 3 = 0$, $x, y \in \mathbb{R}$, then x and y respectively lie in the intervals

(1) $[-\frac{1}{3}, \frac{1}{3}]$ and $[-\frac{1}{3}, \frac{1}{3}]$

(2) $[1, 3]$ and $[-\frac{1}{3}, \frac{1}{3}]$

(3) $[-\frac{1}{3}, \frac{1}{3}]$ and $[1, 3]$

(4) $[1, 3]$ and $[1, 3]$

Q62. If $S = \{z \in \mathbb{C} : \frac{z-i}{z+2i} \in \mathbb{R}\}$, then

(1) S is a circle in the complex plane

(2) S contains exactly two elements

(3) S contains only one element

(4) S is a straight line in the complex plane

Q63. If for $x, y \in \mathbb{R}$, $x > 0$, $y = \log_{10} x + \log_{10} x^{1/3} + \log_{10} x^{1/9} + \dots$ upto ∞ terms and $\frac{2+4+6+\dots+2y}{3+6+9+\dots+3y} = \frac{4}{\log_{10} x}$, then the ordered pair (x, y) is equal to

(1) $(10^6, 6)$

(2) $(10^6, 9)$

(3) $(10^2, 3)$

(4) $(10^4, 6)$

Q64. If $0 < x < 1$, then $\frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 + \dots$, is equal to

(1) $x\left(\frac{x+1}{1-x}\right) + \log_e(1-x)$

(2) $x\left(\frac{1-x}{1+x}\right) + \log_e(1-x)$

(3) $\frac{1+x}{1-x} + \log_e(1-x)$

(4) $\frac{1-x}{1+x} + \log_e(1-x)$

Q65. $\sum_{k=0}^{20} \binom{20}{k}^2$ is equal to

(1) $^{40}C_{21}$

(2) $^{41}C_{20}$

(3) $^{40}C_{20}$

(4) $^{40}C_{19}$

Q66. Let A be a fixed point $(0, 6)$ and B be a moving point $(2t, 0)$. Let M be the mid-point of AB and the perpendicular bisector of AB meets the y -axis at C . The locus of the mid-point P of MC is

(1) $3x^2 + 2y - 6 = 0$

(2) $2x^2 - 3y + 9 = 0$

(3) $3x^2 - 2y - 6 = 0$

(4) $2x^2 + 3y - 9 = 0$

Q67. A tangent and a normal are drawn at the point $P(2, -4)$ on the parabola $y^2 = 8x$, which meet the directrix of the parabola at the points A and B respectively. If $Q(a, b)$ is a point such that $AQBP$ is a square, then $2a + b$ is equal to

(1) -12

(2) -20

(3) -16

(4) -18

Q68. If α, β are the distinct roots of $x^2 + bx + c = 0$, then $\lim_{x \rightarrow \beta} \frac{e^{2(x^2+bx+c)} - 1 - 2(x^2+bx+c)}{(x-\beta)^2}$ is equal to

(1) $2(b^2 + 4c)$

(2) $b^2 - 4c$

(3) $2(b^2 - 4c)$

(4) $b^2 + 4c$

Q69. The statement $(p \wedge (p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow r$ is

(1) a tautology

(2) equivalent to $q \rightarrow \sim r$

(3) a fallacy

(4) equivalent to $p \rightarrow \sim r$

Question Paper

Q70. Let $\frac{\sin A}{\sin B} = \frac{\sin(A-C)}{\sin(C-B)}$, where A, B, C are angles of a triangle ABC . If the lengths of the sides opposite these angles are a, b, c respectively, then

- (1) b^2, c^2, a^2 are in A.P. (2) c^2, a^2, b^2 are in A.P.
 (3) $b^2 - a^2 = a^2 + c^2$ (4) a^2, b^2, c^2 are in A.P.

Q71. If the matrix $A = \begin{bmatrix} 0 & 2 \\ K & -1 \end{bmatrix}$ satisfies $A(A^3 + 3I) = 2I$, then the value of K is

- (1) $\frac{1}{2}$ (2) 1
 (3) -1 (4) $-\frac{1}{2}$

Q72. If $(\sin^{-1} x)^2 - (\cos^{-1} x)^2 = a$; $0 < x < 1$, $a \neq 0$, then the value of $2x^2 - 1$ is

- (1) $\cos\left(\frac{2a}{\pi}\right)$ (2) $\sin\left(\frac{2a}{\pi}\right)$
 (3) $\cos\left(\frac{4a}{\pi}\right)$ (4) $\sin\left(\frac{4a}{\pi}\right)$

Q73. A wire of length 20 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into a regular hexagon. Then the length of the side (in meters) of the hexagon, so that the combined area of the square and the hexagon is minimum, is

- (1) $\frac{10}{2+3\sqrt{3}}$ (2) $\frac{5}{3+\sqrt{3}}$
 (3) $\frac{10}{3+2\sqrt{3}}$ (4) $\frac{5}{2+\sqrt{3}}$

Q74. If $U_n = \left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \dots \left(1 + \frac{n^2}{n^2}\right)^n$, then $\lim_{n \rightarrow \infty} (U_n)^{\frac{-4}{n^2}}$ is equal to

- (1) $\frac{16}{e^2}$ (2) $\frac{4}{e}$
 (3) $\frac{4}{e^2}$ (4) $\frac{e^2}{16}$

Q75. $\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to

- (1) 5 (2) 10
 (3) 8 (4) 6

Q76. Let us consider a curve, $y = f(x)$ passing through the point $(-2, 2)$ and the slope of the tangent to the curve at any point $(x, f(x))$ is given by $f(x) + xf'(x) = x^2$. Then

- (1) $x^3 - 3xf(x) - 4 = 0$ (2) $x^2 + 2xf(x) - 12 = 0$
 (3) $x^3 + xf(x) + 12 = 0$ (4) $x^2 + 2xf(x) + 4 = 0$

Q77. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = 2(y + 2 \sin x - 5)x - 2 \cos x$ such that $y(0) = 7$. Then $y(\pi)$ is equal to

- (1) $7e^{\pi^2} + 5$ (2) $e^{\pi^2} + 5$
 (3) $2e^{\pi^2} + 5$ (4) $3e^{\pi^2} + 5$

Q78. The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to a line, whose direction ratios are 2, 3, -6 , is

- (1) 2 (2) 5
 (3) 3 (4) 1

Q79. Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ units from the origin, which contains the line of intersection of the planes $x - y - z - 1 = 0$ and $2x + y - 3z + 4 = 0$, is

- (1) $-x + 2y + 2z - 3 = 0$ (2) $3x - 4z + 3 = 0$
 (3) $3x - 1y - 5z + 2 = 0$ (4) $4x - y - 5z + 2 = 0$

Q80. When a certain biased die is rolled, a particular face occurs with probability $\frac{1}{6} - x$ and its opposite face occurs with probability $\frac{1}{6} + x$. All other faces occur with probability $\frac{1}{6}$.

Note that opposite faces sum to 7 in any die. If $0 < x < \frac{1}{6}$, and the probability of obtaining total sum = 7, when such a die is rolled twice, is $\frac{13}{96}$, then the value of x is

Question Paper

$$(1) \frac{1}{16}$$

$$(3) \frac{1}{8}$$

$$(2) \frac{1}{12}$$

$$(4) \frac{1}{9}$$

Q81. If $A = \{x \in R : |x - 2| > 1\}$, $B = \{x \in R : \sqrt{x^2 - 3} > 1\}$, $C = \{x \in R : |x - 4| \geq 2\}$ and Z is the set of all integers, then the number of subsets of the set $(A \cap B \cap C)^c \cap Z$ is _____.

Q82. A number is called a palindrome if it reads the same backward as well as forward. For example 285582 is a six digit palindrome. The number of six digit palindromes, which are divisible by 55, is _____.

Q83. Let the equation $x^2 + y^2 + px + (1 - p)y + 5 = 0$ represent circles of varying radius $r \in (0, 5]$. Then the number of elements in the set $S = \{q : q = p^2 \text{ and } q \text{ is an integer}\}$ is _____.

Q84. If the minimum area of the triangle formed by a tangent to the ellipse $\frac{x^2}{b^2} + \frac{y^2}{4a^2} = 1$ and the co-ordinate axis is kab , then k is equal to _____.

Q85. Let n be an odd natural number such that the variance of $1, 2, 3, 4, \dots, n$ is 14. Then n is equal to _____.

Q86. If the system of linear equations

$$2x + y - z = 3$$

$$x - y - z = \alpha$$

$$3x + 3y + \beta z = 3$$

has infinitely many solutions, then $|\alpha + \beta - \alpha\beta|$ is equal to _____.

Q87. If $y^{1/4} + y^{-1/4} = 2x$, and $(x^2 - 1)\frac{d^2y}{dx^2} + \alpha x \frac{dy}{dx} + \beta y = 0$, then $|\alpha - \beta|$ is equal to _____.

Q88. The number of distinct real roots of the equation $3x^4 + 4x^3 - 12x^2 + 4 = 0$ is _____.

Q89. If $\int \frac{dx}{(x^2 + x + 1)^2} = a \tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + b\left(\frac{2x+1}{x^2+x+1}\right) + C$, $x > 0$ where C is the constant of integration, then the value of $9(\sqrt{3}a + b)$ is equal to _____.

Q90. Let $\vec{a} = \hat{i} + 5\hat{j} + \alpha\hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + \beta\hat{k}$ and $\vec{c} = -\hat{i} + 2\hat{j} - 3\hat{k}$ be three vectors such that, $|\vec{b} \times \vec{c}| = 5\sqrt{3}$ and $\vec{a}$ is perpendicular to $\vec{b}$. Then the greatest amongst the values of $|\vec{a}|^2$ is _____.

ANSWER KEYS

1. (2)	2. (2)	3. (3)	4. (2)	5. (2)	6. (2)	7. (4)	8. (2)
9. (4)	10. (3)	11. (2)	12. (1)	13. (2)	14. (3)	15. (4)	16. (2)
17. (1)	18. (3)	19. (1)	20. (3)	21. (12)	22. (2)	23. (2)	24. (2)
25. (1210)	26. (20)	27. (3)	28. (11)	29. (5)	30. (224)	31. (3)	32. (3)
33. (4)	34. (1)	35. (4)	36. (2)	37. (4)	38. (3)	39. (4)	40. (4)
41. (3)	42. (4)	43. (3)	44. (2)	45. (2)	46. (2)	47. (1)	48. (3)
49. (1)	50. (1)	51. (4)	52. (518)	53. (18)	54. (4)	55. (316)	56. (1125)
57. (82)	58. (3155)	59. (40)	60. (2)	61. (2)	62. (4)	63. (2)	64. (1)
65. (3)	66. (4)	67. (3)	68. (3)	69. (1)	70. (1)	71. (1)	72. (2)
73. (3)	74. (4)	75. (1)	76. (1)	77. (3)	78. (4)	79. (4)	80. (3)
81. (256)	82. (100)	83. (61)	84. (2)	85. (13)	86. (7)	87. (17)	88. (4)
89. (15)	90. (90)						